

Benchmarking continuous optimization techniques for DAG learning

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Abstract

Learning Bayesian Network (DAG) structures is computationally challenging due to acyclicity constraints. We compare continuous optimization methods (NOTEARS, DAG-GNN) against classical algorithms (PC, FGES, Hill Climbing) using synthetic (5-37 nodes) and real-world biological datasets. Results reveal a distinct trade-off: continuous methods scale better with network complexity, achieving lower Structural Hamming Distances, whereas classical algorithms scale better with dataset size but bottleneck on large networks. Notably, the graph-based DAG-GNN underperformed the simpler NOTEARS baseline in both accuracy and efficiency.

Introduction

Domain

Learning a Bayesian Network from data can be a difficult task. The space of DAGs over a set of nodes is a combinatorial space and traditional approaches have a super-exponential worst case complexity, mainly owed to the acyclicity constraint to be enforced on the learned DAG. Continuous numerical optimization techniques can be a viable option to speed up the search.

Aim

The objective of this study is to evaluate the performance of numerical optimization algorithms, compared to a baseline of classical algorithm such as PC (Spirtes and Glymour 1991), FGES (Ramsey et al. 2017), and hill climbing (Heckerman, Geiger, and Chickering 1995). We seek to understand how good the reconstructed DAG is and the compute resources needed to reconstruct it. We also want to understand the scaling of these resources as the number of variables grows or the number of data points grows.

Method

The bnlearn repository contains a repository of prebuilt Bayesian networks that can be used to generate a synthetic dataset, with DAGs ranging from just 5 variables to 1041. We selected 3 of them in the range of variables 5-37. To avoid only using synthetic data we also tested on the protein signaling work introduced in (Sachs et al. 2005) which

contains a known ground truth DAG with 11 nodes and a dataset of 11672 real samples. The full experiment covers 4 networks with node count ranging from 5 to 37.

We investigate the NOTEARS algorithm (Zheng et al. 2018) and the DAG-GNN algorithm (Yu et al. 2019). NOTEARS was the first work to transform DAG structure learning from a combinatorial search into a continuous optimization problem, by introducing an algebraic characterization of the acyclicity constraint which allows the network structure to be estimated using black-box numerical solvers. DAG-GNN was then chosen as an extension. It works by employing a GNN-parameterized Variational Autoencoder (VAE) to capture the underlying data distribution, which seemed a natural fit to learn the DAG structure of the probability distribution.

For each experiment, we track Structural Hamming Distance (SHD) between the ground truth DAG and the reconstructed one together with the execution time of the algorithm.

Results

NOTEARS achieved the highest overall performance recovering the best DAGs among all of the tested algorithms. DAG-GNN performed worse than NOTEARS, while being much more costly to run. These approaches are not significantly affected by an increase in variable count, while being much more sensitive to sample count in the dataset used, a limitation not observed in classical methods.

Model

Many of the Bayesian networks tested for this study come from the bnlearn repository of networks. Specifically, we used the CANCER composed of 5 variables, the CHILD composed of 20 variables, and the ALARM one composed of 37 variables. All of these networks include discrete variables only. For more details on the networks we refer the reader to the bnlearn repository.

In addition to these discrete benchmarks, we evaluate algorithms on a real-world continuous dataset introduced by Sachs et al. (Sachs et al. 2005), derived from single-cell flow cytometry measurements. The ground truth network consists of 11 continuous variables representing signaling proteins.

Analysis

Experimental setup

NOTEARS, DAG-GNN, and FGES are run using the implementations provided by the gCastle Python toolbox, while PC and Hill Climbing are run using the implementations provided by the pgmpy library.

Each algorithm has its own hyper-parameters so first we run a hyper-parameter grid search over each dataset for each algorithm to derive good hyper-parameter values for each dataset-algorithm combination.

Then, we execute the algorithm on each dataset and compare the learned DAG against the ground-truth one computing graph reconstruction metrics, along with execution time of the algorithm. The most important metric is SHD, which measures how many edge additions, deletions, or inversions need to be performed on a graph to transform it into the other (lower means better).

For all the networks from the bnlearn repository we generated a synthetic dataset of 1000 samples, while for the dataset from (Sachs et al. 2005) we used all 11672 samples as a higher data regime. We also note that in the preliminary phase the BARLEY graph (48 nodes) from the bnlearn repository was tested, but ultimately discarded because all of the classical algorithms took extremely long to train on it. Experiment code is available on Github.¹

Results

On the very small CANCER NOTEARS and DAG-GNN do not appear to be helpful compared to classical algorithms as they both recover DAGs with worse SHD than other methods achieved, where fGES is the clear winner. As the number of nodes increases, NOTEARS starts showing an edge, achieving lower SHD than any other method. Notably, DAG-GNN does somewhat better than classical algorithms, but does not beat NOTEARS. It is still believable that it could beat NOTEARS as the network becomes bigger and the relationships between variables more complex, but this investigation is left as future work. Finally we note that on the biggest network, PC was the winner, beating NOTEARS by 5 points.

For execution time, we observe that all algorithms take longer with more variables introduced in the network, with fGES failing to terminate on ALARM and, together PC, were not applicable to the BARLEY network (48 nodes) leading to its removal from the experiment. On the other hand, their scaling w.r.t. the dataset size is much less favourable for numerical algorithms: Hill climbing, PC, and FGES all take roughly the same amount of time on the Sachs et al. dataset that it took them on the CHILD one; NOTEARS instead took roughly 20 times more time to terminate and DAG-GNN took 16 times.

Table 1 contains the performance of each algorithm across all datasets.

Table 1: nSHD is the average "scale normalized" SHD, the SHD divided by the number of edges in the original DAG

Algorithm	Avg. Time (s)	nSHD	CANCER	Sachs	CHILD	ALARM
NOTEARS	44.22	0.905	0.250	0.451	0.293	0.348
DAG-GNN	519.09	0.993	0.112	0.195	0.137	0.161
HC	9.62	1.282	0.215	0.237	0.333	0.274
PC	3.17	0.928	0.374	0.477	0.442	0.455
FGES	3.66	0.877	0.452	0.512	0.523	0.514

Conclusion

In conclusion, this study reveals a trade-off: continuous methods yield better structural accuracy for complex networks, while classical algorithms scale better with larger datasets. Surprisingly, DAG-GNN underperformed against simpler baselines. Current limitations include classical bottlenecks on very large networks and continuous solvers' sensitivity to large samples. Future work should address these issues by optimizing continuous methods for larger datasets and evaluating advanced models on more complex graphs.

Links to external resources

- pgmpy library: <https://pgmpy.org/index.html>
- gCastle library: <https://github.com/huawei-noah/trustworthyAI/blob/master/gcastle/README.md>
- bnlearn repository: <https://www.bnlearn.com/bnrepository/>

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¹<https://github.com/Tarekun/bayesian-continuous>